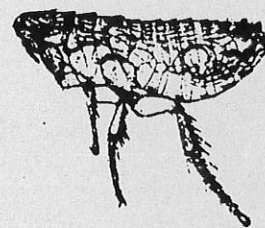


flea NEWS 43



H. E. Krampitz

Department of Entomology Iowa State University Ames, Iowa 50011-3222

EDITORIAL

Flea News is a twice-annual newsletter begun in 1973 by Frans Smit, then Custodian of the Rothschild Collection of Fleas. In 1980, when Mr. Smit retired, its continuation was assumed by Robert & Joanne Lewis, with financial support of the Department of Entomology at Iowa State University. As a result, it has been distributed, free of charge, to roughly 175-200 persons and institutions in over 35 countries for the past eleven years. Approximately half of the recipients have been from the United States. Over the years many recipients have offered financial assistance. Although we very much appreciated the offers, we have always refused because of the problems associated with bookkeeping and foreign exchange. Our Governor, the Legislature and the Board of Regents have now managed to change all of this. The result: We are searching for outside financial support to continue the newsletter. Fortunately, **ZOECON**, a **Sandoz** company headquartered in Dallas, Texas, has generously agreed to support the production of this issue and the two issues to be produced next year. After that, who knows? We have contacted recipients employed by other firms which produce and/or market products used in flea control but we have so far received no response. With the June, 1992 issue we intend to update our mailing list. By then we should know if any other extramural support will be forth-

Contents

Announcements 426
Editorial 426
Krampitz Obituary 431-433
Literature 435-443
Mailing List Changes 431
New Book 428
Surpluss Reprints 428-431
Taxon Lists 433-435

coming. If not, we will have to request support from recipients. Without it, **Flea News** will be discontinued after the December 1992 issue.

ANNOUNCEMENTS

The 4th National Conference on Urban Entomology will be held 23-26 February 1992 at the Adult Education Center, University of Maryland, College Park, MD. In addition to sections dealing with termites and cockroaches, there will be a section on fleas *et al.* moderated by Drs Michael Dryden and Phil Koehler, Veterinary Pathobiology, Kansas State University, Manhattan, KS. The registration fee of \$85.00 covers 2 luncheons, 1 dinner and a copy of the Proceedings.

For further information, contact Dr. Judith Bertholf, DowElanco, 13355 Noel Road, Suite 1045, Dallas, TX 75240-6605 or call (214) 702 1371.

D E C E M B E R

1991

Announcements (Continued)

The following presentations dealing with fleas were to be delivered at the 23rd Annual Conference of the Society for Vector Ecology, Reno (Sparks), NV, November 17-20, 1991.

Reproductive activity of *Synosternus cleopatrae* (Siphonaptera) in relation to host factors. T. Lehmann, University of Arizona, Tucson, AZ.

Estimation of genetic variation in populations of the rock squirrel flea, *Oropsylla [(Diamanus)] montana*, by allozyme analysis. M. Quintana & D. Howard, New Mexico State University, Las Cruces, NM.

Ivermectin and its effects on blood-feeding arthropods: As review and prospective analysis. R. G. Endris, Merck, Sharp & Dohme, Rahway, NJ.

Nancy C. Hinkle, a Ph.D. candidate at the University of Florida has been named the Southeastern Branch recipient of the Comstock Award of the Entomological Society of America. Her dissertation focuses on the estimation of flea populations on the host and the use of insect growth regulators to suppress populations. Her research developed proper use patterns designed to reduce insecticide exposure and hazard to pets and humans. In 1991 she was presented the University of Florida President's Award as "Outstanding Student". She has also been named to "Outstanding Young Women of America". Hinkle has served as moderator for sessions in subjects ranging from Urban Entomology to Toxicology at national and branch meetings of the ESA. She received both her B. S. and M. S. degrees from Auburn University. (Excerpted from the ESA Newsletter 14(11): 5.)

Nancy Briscoe, an undergraduate student in the Department of Biological Sciences, Marshall University, Huntington, WV, presented the following paper at the Second Student Presentation Competition, sponsored by the Helminthological Society of Washington at the 618th meeting of the Society on 13 March 1991. "Population dynamics of *Orchopeas leucopus* (Siphonaptera: Ceratophyllidae) and *Epitedia wenmanni* (Siphonaptera: Hystrihopsyllidae) from the white-footed mouse, *Peromyscus leucopus*, in Mason Co., West Virginia. (Excerpted from J. Helminthol. Soc. Wash. 58(2): 201.)

Bird-Parasite Interactions, edited by Jenella E. Loye and M. Zuk, is now available from the Oxford University Press. The editors have drawn together a comprehensive range of specialists in the field to provide a reference to the current work on bird-parasite interactions. Divided into four parts, the introduction provides the contemporary context of research in bird-parasite systems and presents case studies which combine theoretical concepts with experimental data. Subsequent parts focus on the ecology, behavioural responses and the effect of parasitism on sexual selection. *Oxford Ornithological Series*. 0-19-857738-9, 406 pp., 54 figures, May 1991, £37.50.

Sections of potential interest to readers of Flea News are:

Current theory of host-parasite interaction - C. A. Toft.

The effects of ectoparasites on cliff swallow growth and survival - C. Delanney & A. Cruz.

The nest protection hypothesis: the adaptive use of plant secondary components by European starlings - J. E. Loye & S. P. Carroll.

Nest ectoparasite abundance and cliff swallow colony site selection, nestling development and departure time - D. C. Duffy.

Coevolution of avian grooming and ectoparasite avoidance - M. D. Saumier, M. E. Rau & D. M. Bird.

Publications Available. The following papers are excess duplicates in the National Museums reprint files. In an effort to have this research information in the hands of researchers, we are offering these papers free of charge, but ask that you send us copies of your research publications in return. They are offered on a first come first serve basis. If interested, contact Ms. Nancy Adams, Department of Entomology, National Museum of Natural History, Smithsonian Institution, MRC 105, Washington, DC 20560. Numbers in () indicate number of copies.

Anoplura

P. T. Johnson.

1957. The identity of *Pediculus spiculifer* Gervais. (204)

1957. Descriptions of two new species of *Eulinognathus* Cummings & redescription of *E. aculeatus* (Neumann). (20)

1958. Type specimens of lice (Anoplura) in the United States National Museum. (35)

1958. A new species of Anoplura from the Philippines. (33)

1959. The rodent-infesting Anoplura (Sucking lice) of Thailand, with some remarks on some related species. (9)

- 1960. A new species of *Hoplopleura* from Australia (Hoplopleuridae). (121)
- 1960. The sucking lice (Anoplura) of Egypt. I. Species infesting rodents. (1)
- 1960. The Anoplura of African rodents and insectivores. (2)
- 1962. Redescriptions of two cervid-infesting Anoplura from Southeast Asia. (53)
- 1962. Notes and descriptions of African lice (Anoplura). (7)
- 1962. Three new Anoplura from African rodents (Anoplura: Hoplopleuridae). (39)
- 1962. The species of *Fahrenholzia* Kellogg & Ferris from spiny pocket mice (Anoplura: Hoplopleuridae). (39)
- 1963. Two rare Anoplura from Kenya. (66)
- 1964. The hoplopleurid lice of the Indo-Malayan Subregion (Anoplura). (33)
- 1969. *Hamophthirus galeopithecii* Mjoberg rediscovered; with the description of a new family of sucking lice (Hamophthiriidae). (72)
- Scanlon, J. E. & P. T. Johnson.
1957. On some microtine-infesting *Polyplax* (Anoplura). (45)
- Wenzel, R. L. & P. T. Johnson.
1966. Checklist of the sucking lice of Panama. (Anoplura) (67)

Mallophaga

M. A. Carriker.

- 1944. Studies in Neotropical Mallophaga. (III). (5)
- 1949. On a collection of Mallophaga from Guam, Marianas Islands. (60)
- 1949. Some bird lice of the genera *Acidoproctus* and *Quadriceps*. (35)
- 1954. Studies on Neotropical Mallophaga. (XI). (12)
- 1957. Studies on Neotropical Mallophaga. (XVI). (8)
- 1960. Studies on Neotropical Mallophaga. (XVII). (9)
- 1963. New and little known Mallophaga from Venezuelan birds. (Part II). (25)
- 1966. A revision of the genus *Furnaricola* (Mallophaga) with descriptions of new species. (24)

R. E. Elbel.

- 1967. Amblyceran Mallophaga found on the Bucerotidae (Hornbills). (28)

K. C. Emerson.

- 1960. Two new species of *Craspedorrhynchus* (Mallophaga) from N. America. (5)
- 1961. Three new species of Mallophaga from the great horned owl. (5)
- 1962. Mallophaga (chewing lice) occurring on the turkey. (29)
- 1962. The families and genera of North American (N. of Mexico) Mallophaga. (17)
- 1962. List of Mallophaga found on North American (N. of Mexico) mammals. (19)
- 1962. Mallophaga found on domestic fowl. (7)
- 1964. Checklist of the Mallophaga of N. America (N of Mexico) Pt. II. Amblycera. (1)

1964. Checklist of the Mallophaga of N. America (N of Mexico) Pt. I. Ischnocera. (3)
1967. Carriker on Mallophaga. (40)
1969. New species and records of Mallophaga (Trichodectidae) from Nigerian mammals. (2)
1972. Checklist of the Mallophaga of N. America (N of Mexico) Pt. I. Ischnocera. (1)
- K. C. Emerson & C. J. Stojanovich.
1964. A new species of Mallophaga from the Mikado pheasant. (65)
- K. C. Emerson & R. D. Price.
1968. A new species of *Parafelicola* (Trichodectidae) from Mozambique. (2)
- H. E. Ewing.
1930. The taxonomy and host relationships of the biting lice of the genera *Dennyus* and *Eureum*... (3)
- K. P. Hajela & B. K. Tandan.
1970. Species of *Ardeicola* (Mallophaga) parasitic on birds of the family Threskiornithidae. (4)
- P. Kumar & B. K. Tandan.
1971. The species of *Ardeicola* (Phthiraptera: Ischnocera) parasitic on Ciconiidae. (15)
- R. A. Ward.
1955. Biting lice of the genus *Saemundssonina* (Mallophaga: Philopteridae) occurring on terns. (23)

Siphonaptera

- ✓ = Have
- write
- need
- P. T. Johnson.
- ✓ 1954. *Myodopsylla setosa* and *Tiarapsylla bella*, new species of fleas from Peru. (70)
1954. Notes on *Pleochaetis* Jordan, 1933, from Colombia, with the description of a new species. (5) *J. Wash. Acad. Sci.* 44(9): 289-96.
- ✓ 1955. The genus *Chaetopsylla* Kohaut, 1903, in North America, with the description of a new species. (93)
- ✓ 1957. A redescription of the flea genus *Libyastus* Jordan, 1936. with the description of a new species. (90)
- ✓ 1961. A revision of the species of *Monopsyllus* Kolenati in North America. (14)
- P. T. Johnson & J. N. Layne.
- ✓ 1961. A new species of *Polygenis* Jordan from Florida, with remarks on its host relationships & zoogeographic significance. (37)
- J. Segerman.
- ✓ 1970. Description of *Lagaropsylla setzeri* nov. spec. and the male of *Stivalius nigeriensis* Jordan from the Ethiopian Region. (4)

- ✓ 1971. A new species of *Ctenophthalmus* from Ghana. (10)

Misc.

G. E. Watson & A. B. Amerson, Jr.

1967. Instructions for collecting bird parasites. (200)

The following colleagues have sent reprints since Flea News 42. Thank you. We very much appreciate your cooperation!

K. Bartkowska, J.-C. Beaucournu, D. Cyprich, A. Dudich, L. Durden, A. I. Goncharov, J. E. Llorente-Bousquets, S. G. Medvedev, E. Méndez, M. Rothschild, V. F. Sapegina, V. S. Vashchenok, Wang D.-q. & Xie B.-q.

Mailing List Changes.

Dr. Mathias Kiefer, Westendstr. 79,8000, München 2, GERMANY.

Dr. Michael B. Usher, Chief Scientific Advisor, Nature Conservancy Council for Scotland, 2/5 Anderson Place, Edinburgh EH6 5NP, SCOTLAND.

◆◆◆

HEINZ EBERHARD KRAMPITZ



Prof. Dr. med. Heinz E. Krampitz was born in Breslau, Lower Silesia, Prussia (now Wrocław, Poland), on 2 July 1921, and died in Munich, Germany, on 12 May 1991, at the age of 69. His father was an otolaryngologist. He attended the Humanistisches Gymnasium in Breslau. In 1941 he was inducted into the Medical Service of the German Army and saw active duty during the siege of Stalingrad where he was seriously wounded. He was to spend several of the following years in and out of hospital, recovering from the effects of these wounds. During the "Festungzeit" in Breslau in 1945 he worked in the military hospital there, ultimately becoming Director of the laboratory. After escaping to the west in 1946 he resumed his academic

studies, first in Mainz and later in Frankfurt am Main, where he sat for the state examination in 1948. 1951 saw him conducting research in southern Italy, Sicily and the French Mediterranean coast. For the next nine years he worked at the State Hygiene Institute and at the University of Frankfurt. From 1960 to 1963 he attended courses in Tropical Medicine and Medical Parasitology, and took the diploma examination at the Hamburg Tropical Institute, where he was a scientific collaborator in the Department of Tropical Hygiene, Protozoology and Veterinary Medicine. During this time he also lectured in courses on Tropical Medicine. Part of this time was spent studying the parasitic protozoa and malaria in cooperation with the Institute of Hygiene at Palermo University. In 1963 he was made Scientific Assistant in the Institute for Infections and Tropical Medicine of the Universität München, where he remained until the end of his scientific career. He was made "Privatdozent" [Lecturer] in 1970 and Professor in 1980.

In addition to his teaching and research activities at the University, Professor Krampitz was active in a number of professional organizations and activities. During part of 1965-1966, he served as Visiting Research Officer to the East African Trypanosomiasis Research Organization in Tororo, Uganda. In 1973 he was a participant in the German Speaking Work Group for the Standardization of Medical Terminology of C. I. O. M. S. / W. H. O. (Heidelberg) and was active in organizing the III International Congress of Parasitology (ICOPA III) which was held in Munich in September of 1974. It was at this Congress that I had the pleasure of making his acquaintance.

Dr. Krampitz was a specialist in Parasitology, Pathobiology and Epidemiology, and most of his studies dealt with protozoans and related organisms. Still, of the 156 scientific publications authored or co-authored by him, nine of them involve fleas in some manner. A list of these is included below.

Information upon which this account is based was provided by Mrs. Krampitz, transmitted through the kind offices of Dr. Mathias Kiefer, both of whom we thank.

-R.E.L.

Krampitz publications dealing with fleas

1965. Beobachtungen an einer Laboratoriumszucht von *Leptopsylla segnis* Schönherr,

- 1811 (Insecta, Siphonaptera). Z. Parasitenk. 26: 197-214.
1967. A survey of fleas and their hosts in Tororo, Uganda. East Afr. Trypanosomiasis Res. Org. Ann. Rept., Nairobi 1966, p. 94.
1968. Felduntersuchungen zur Verbreitung der Flohfauna in einer Kulturlandschaft Ost-Ugandas mit besonderer Beachtung des Menschen. Z. Tropenmed. Parasitol. 19: 297-305.
1976. Flöhe als Objekte biologischer Forschung (Kurzberichte aus der Wissenschaft). Naturw. Rsch. 29: 14.
1978. Investigations into the blood uptake of larval fleas. Short Comm. IV Internat. Congr. Parasitol., Warschau 1978, Sect. C: 216-217.
1979. Flohverbreitung, Geomedizin und Erdgeschichte. Gedanke anlässlich der 100. Wiederkehr des Geburtstages von Charles Nathaniel Rothschild am 9. Mai 1977. In: H.-J. Juszat, *Geomedizin in Forschung und Lehre*. Steiner Verl., Wiesbaden. pp. 117-122.
1980. with V. Wongchari. The development of *Hepatozoon erhardovae* in experimental mammalian and arthropod hosts. 1. The evaluation of suitable vectors. In: R. Traub & H. Starke (Eds.) *Fleas*. (Proc. Internat. Conf. on Fleas, Ashton Wold, 1977). Balkema. Rotterdam. pp. 349-358.
1980. Host preference, sessility and mating behaviour of *Leptopsylla segnis* reared in captivity. *Ibid.* pp. 371-378.
1981. Development of *Hepatozoon erhardovae* Krampitz, 1964 (Protozoa: Haemogregarinidae) in experimental mammalian and arthropod hosts. 2. Sexual development in fleas and sporozoite indices in xenodiagnosis. Trans. R. Soc. Trop. Med. Hyg. 75: 155-157.



A PREFATORY NOTE TO THE SPECIES LIST -- In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those taxa described in 1989. Readers are urged to call any omissions in these lists to our attention.

- 98901 Palaeopsylla laxidigita Xie & Gong
 98902 Palaeopsylla polyspina Xie & Gong
 98903 Palaeopsylla medimina Xie & Gong
 98904 Ctenophthalmus (Palaeoctenophthalmus) harputus Aktaş
 98905 Ctenophthalmus (Ctenophthalmus) solutus nasutus Beaucournu, Valle & Launay
 98906 Citellophilus simplex rosickyi Cyprich
 98907 Xenopsylla guancha Beaucournu, Alcover & Launay

- 98908 *Stenischia chini* Xie & Lin
- 98909 *Stenischia liae* Xie & Lin
- 98910 *Stenischia liui* Xie & Lin
- 98911 *Stenischia wui* Xie & Lin
- 98912 *Jellisonia amadoi* Ponce-Ulloa
- 98913 *Jellisonia mexicana* Ponce-Ulloa
- 98914 *Plocopsylla crypta* Beaucournu & Gallardo
- 98915 *Rowleyella gongshanensis* Gong & Duan

Provisional lists of taxa described since 1990

- Hectopsylla pascuali* Beaucournu & Alcover
- Plocopsylla consobrina* Beaucournu & Alcover
- Agastopsylla boxi gibbosa* Beaucournu & Alcover
- Chiliopsylla allophyla tonnii* Beaucournu & Alcover
- Geusibia digitiforma* Gong & Lin
- Rhadinopsylla* (*Sinorhadinopsylla*) *leii* Xie, Gong & Duan
- Lagaropsylla duodecima* Beaucournu & Kock
- Ceratophyllus* (*Emmareus*) *liae* Wu & Li
- Smitipsylla quadrata* Xie & Li
- Rowleyella nujiangensis* Lin & Xie
- Macrostylophora bispiniforma gongshanensis* Gong & Xie
- Hystrichopsylla llorentei* Ayala & Morales
- Jellisonia guerrerensis* Morales
- Ctenoparia intermedia* Beaucournu & Kelt
- Plocopsylla wilesi* Beaucournu & Kelt
- Plocopsylla silewi* Beaucournu & Kelt
- Tiamastus gallardoi* Beaucournu & Kelt
- Liuopsylla clavula* Xie & Duan
- Xenopsylla trispinus tenuis* Beaucournu & Kock
- Nosopsyllus wualis rongjiangensis* Li & Huang
- Nosopsyllus wualis bosensis* Li & Pan
- Nosopsyllus wualis diandongensis* Li, Xie & Yang
- Nosopsyllus elongatus longchuanensis* Li, Xie & Wu
- Nosopsyllus elongatus puerensis* Li, Xie & Yu
- Nosopsyllus elongatus yanshanensis* Li, Xie & Liao
- Coptopsylla lamellifer tarimensis* Yu, Ye & Cai
- Vermipsylla yeae* Yu & Li
- Stenoponia meridiana* Cao, Xie & Yu

Ctenophyllus (Ctenophyllus) armatus altaicus Yu
Ophthalmopsylla (Ophthalmopsylla) volgensis wuqiaensis Yu, Shen & Ye
Callopsylla (Callopsylla) yui Ye & Jiang
Leptopsylla (Leptopsylla) algira vogeli Beaucournu
Ischnopsyllus (Ischnopsyllus) kilitbahiricus Aktaş

1991

Xenopsylla cuisancei Beaucournu & Morel
Parapsyllus senellarti Beaucournu & Rodhain
Ctenophthalmus (Ethioctenophthalmus) evidens abyssinicus Beaucournu
Xiphiopsylla poggii Beaucournu
Neopsylla biseta bijiangensis Xie, Fang & Hu
Doratopsylla jii Xie & Tian
Doratopsylla liui Xie & Xie
Doratopsylla coreana jianchuanensis Xie & Yang
Chaetopsylla (Chaetopsylla) ailuropodae Jeu, Wang & Li
Nycteridopsylla singula Rybin

✦✦✦★★★✦✦✦

LITERATURE ON SIPHONAPTERA

(1984-1987)

1984 (List 13)

Anderson, J. F. & L. A. Magnarelli. Avian and mammalian hosts for spirochaete-infected ticks and insects in a Lyme Disease focus in Connecticut. Yale J. biol. Med. 57: 627-641.

1986 (List 11)

Isaäcson, M. Plague and cholera surveillance in southern Africa. S. Afr. med. J. 1986 Supplement, 43-46.

1987 (List 9)

Aktaş, M. Bat fleas of eastern Turkey (east of the Samsun-Iskanderun line). Doğu Turk Zooloji Dergisi 11(3): 111-118.

(1987-1989)

Kamath, V. R. & D. M. Renapurkar. Evaluation of insecticidal activity of propetamphos against common household insect pests. Pesticides 21(12): 27-28.

1988 (List 8)

Magnarelli, L. A. & J. F. Anderson. Ticks and biting insects infected with the etiologic agent of Lyme Disease, *Borrelia burdorferi*. J. clin. Microbiol. 26: 1482-1486.

1989 (List 6)

Abul-Hab, J. & B. A. Shihab. Ectoparasites of some bats from Iraq. Bull. Iraq nat. Hist. Mus. 8: 59-64.

Brasil, D. P., F. G. Carvalho, C. R. Almedia & A. M. P. Almedia. Studies on natural infection with *Yersinia pestis* in fleas from the plague foci in northeast Brazil. Revta Soc. Bras. Med. trop. 22(4): 177-181.

Matias, R. S. Epidemic of tungiasis in Rio Grande do Sul. Revta Soc. Bras. Med. trop. 22(3): 137-142.

Medvedev, S. G. Ecological peculiarities and distribution of the fleas of the family Ischnopsyllidae (Siphonaptera). Parazit. Sb. 36: 21-43.

Medvedev, S. G. Principles and methods of substantiation of higher taxa. Trudy zool. Inst. S. S. S. R. 202: 44-66.

Moriello, K. A. Oh no! Not fleas again! Vet. Focus 1(1): 19...28.

Oparina, O. S., G. S. Starozhitskaya & M. A. Samurov. The ability of fleas from synanthropic rodents and small gerbils to feed on human blood. Med. Parasitol. Parazitar. Bol. (6): 23-25.

Vashchenok, V. S. The ultrastructure of the fat body of female fleas *Xenopsylla cheopis* Roths. (Siphonaptera) in various periods of development of the adult. Parazit. Sb. 36: 83-90, pls. I-IX.

(1989-1990)

Vashchenok, V. S. & G. I. Vashchenok. Survival of *Yersinia pseudotuberculosis* in *Xenopsylla cheopis* Rothsch. (Siphonaptera: Pulicidae). Parazit. Sb. 35: 145-155.

1990 (List 4)

Aktaş, M. Bat fleas of western Turkey (west of the line between Samsun and Iskenderun). Doğu Türk Zooloji Dergisi 14: 107-112.

Aktaş, M. Türkiye den yeni bir yarasa piresi türü *Ischnopsyllus kilitbahiricus* sp. n. (Siphonaptera: Ischnopsyllidae). [A new bat flea species, *Ischnopsyllus kilitbahiricus* sp. n. from Turkey (Insecta: Siphonaptera). Misc. Papers Centre ent. Studies (Ankara) No. 7: 3-7.

Beaucournu, J.-C. Le Professeur Ségué, Parasitologiste. Annls Soc. ent. Fr. (N.S.) 26(3): 317-321.

Beaucournu, J.-C. & D. Kock. Deux puces nouvelles pour Kenya: Description du mâle de *Ctenocephalides chabaudi* Beaucournu & Bain, 1982 et de *Xenopsylla trispinus tenuis* n. subsp. (Insecta: Siphonaptera). Senckenbergiana biol. 70(4/6): 251-260.

Beaucournu, J.-C. & K. Kowalski. Quelques siphonaptères (Insecta) de République démocratique de Corée. Acta zool. cracov. 33(15): 331-344.

Bonge, N. J. Opposing viewpoints on efficacy of ultrasonic flea collars. J. Am. Vet. med. Assoc. 196(9): 1354-1355.

Bryukhanova, G. D., A. K. Akiev, A. P. Beiee & V. K. Mikhailova. The effect of Lindane on the vectorial ability of fleas experimentally infected with plague. Parazitologiya 24(4): 347-349.

Carlson, A. Hedgehogs in veterinary practice. Praktische Tierartz 71(1): 31-35.

Center, S. A., J. F. Randolph, H. N. Erb & S. Reiter. Eosinophilia in the cat: a retrospective study of 321 cases (1975-1986). J. Am. Anim. Hosp. Assoc. 26(4): 349-358.

(1990)

D'Antuono, A., M. Gatti, M. Negosanti, B. Passarini, P. Pauluzzi & M. Reggiana. Tungiasi: uno caso clinico [Tungaisis: a case report] Giorn. Ital. Dermatol. Venereol. 125(6): 259-261.

Kim, K. C., K. C. Emerson & R. Traub. Diversity of parasitic insects: Anoplura, Mallophaga and Siphonaptera. pp. 91-103. In: *Systematics of the North American Insects and Arachnids: Status and Needs*. M. Kosztarab & C. W. Schafer, Eds. Virginia Agricultural Experiment Station Information Series 90-1. Blacksburg: VPI & SU. 247pp.

Lapshov, V. A. & A. P. Inapogi. Data on the contents of the population of Guinea with ectoparasites and rodents obtained by means of a questionnaire. 1. Aims and description of the questionnaire. Med. Parazit. Parazitar Bolz. (1): 23-29.

Ravkin, Yu. S. & V. F. Sapegina. Fleas of the rodents of the southern taiga of Priangarje. Izvest. Sibir. Otdel. Akad. Nauk SSSR. Ser. biol. Nauk 3: 63-68.

Scott, D. W. & M. Paradis. A survey of canine and feline skin disorders seen in a University practice: Small Animal Clinic, University of Montréal, Saint-Hyacinthe, Québec. (1987-1988). Canad. Vet. J. 31(12): 830-835.

Timoshenko, A. F., Yu. V. Mironchuk, N. A. Sukhanov, Yu. D. Ochirov, L. P. Okunev, P. P. Gordeev, N. P. Khutorenko, L. I. Nosoreva, B. V. Shukharbeev & F. I. Khodus. Meadow-field type foci of tularemia in eastern Siberia. Zh. Mikro. Epidem. Immun. 8: 38-42.

Williams, C. K., R. J. Moore & S. J. Robbins. Genetic resistance to myxomatosis in Australian wild rabbits *Oryctolagus cuniculus* (L.) Aust. J. Zool. 38(6): 697-703.

Williams, J. E. Use of ELISA to reveal rodent infection in plague surveillance and control programmes. Bull. Wld Hlth Org. 68(3): 341-345.

1991 (List 2)

Bartkowska, K. Fleas /Siphonaptera/ of small mammals and from their nests of central Roztocze /south-easterly Poland/. Abstract Volume. Fourth European Congress of Entomology. XIII. International Symposium für die Entomofaunistik Mitteleuropas. Gödöllő. No page number (1 page).

Beaucournu, J.-C. Deux Siphonaptères nouveaux d'Ethiopie (Xiphiopsyllidae et Ctenophthalmidae) et essai de clé dichotomique du genre *Xiphiopsylla* Jordan et Rothschild, 1913. Bull. Soc. Fr. Parasit. 9(1): 105-118.

Best, T. L. *Dipodomys nitratoideus*. Mammalian Species 381: 1-7, 5 figs.

Carraway, L. N. & B. J. Verts. *Neotoma fuscipes*. Mammalian Species 386: 1-10, 4 figs.

Carraway, L. N. & B. J. Verts. *Neurotrichus gibbsii*. Mammalian Species 387: 1-7, 3 figs.

Coward, P. S. Fleas in southern England. (Correspondence) Vet. Rec. 129(12): 172.

Cyprich, D. & M. Kiefer. On the occurrence of fleas in the nests of the rat (*Rattus norvegicus* Berkenhout) in Slovakia. Biologia (Bratislava) 46(2): 151-156.

Cyprich, D. & M. Krumpál. Incidence and taxonomy of *Mioctenopsylla arctica arctica* Rothschild, 1922 (Siphonaptera) in the Spitsbergen. Biologia (Bratislava) 46(6): 545-550.

Dudich, A. Die Ektoparasiten kleiner Erdsäugetiere (Insectivora, Rodentia) der Kysucer Beskiden (westlich Karpaten). Siphonaptera. Vlastivedný zborník Považia 16: 203-219.

Dudich, A. Sympatry and parapatry in central European *Peromyscopsylla* species (Siphonaptera: Leptopsyllidae). Abstract Volume. Fourth European Congress of Entomology. XIII. International Symposium für die Entomofaunistik Mitteleuropas. Gödöllő. p. 48.

(1991)

Durden, L. A. A new flea (Siphonaptera) record for Mississippi, with notes on other flea records from the state. J. ent. Sci. 26(3): 310-311.

Fauth, P. T., D. G. Krementz & J. E. Hines. Ectoparasitism and the role of green nesting material in the European starling. Oecologia 88(1): 22-29.

Grove, E., P. Hottel, J. Patton, B. Dodson, P. T. Baietto & B. Kunst. Going to the dogs. The new trend in on-animal products. Pest Management 10(6): 13-14...16.

Hayssen, V. *Dipodomys microps*. Mammalian Species 389: 1-9, 3 figs.

Hinkle, N. C. Flea control: attitudes and experiences of PCOs. Pest Management 10(6): 15.

Hinkle, N. C. A closer look. A super-magnified view of the cat flea's life cycle shows there's more than meets the eye. Pest Management 10(6): 10-11.

Hinkle, N. C., P. G. Koehler, W. H. Kern & R. S. Patterson. Hematophagous strategies of the cat flea (Siphonaptera: Pulicidae). Fla Ent. 74(3): 377-385.

Jeu Ming-hua, Wang Dun-qing & Li Kuei-chen. A new flea of the genus *Chaetopsylla* from the giant panda. Sichuan J. Zool. 10(1): 7-9, figs. 1-3B. [*Chaetopsylla (Chaetopsylla) ailuropodae* Jeu, Wang & Li, 1991]

Kern, W. H., Jr. Cat flea larva - the unknown life stage. Pest Management 10(6): 20-22.

Keymer, I., E. A. Gibson & D. J. Reynolds. Zoonoses and other findings in hedgehogs (*Erinaceus europaeus*): a survey of mortality and review of the literature. Vet. Rec. 128(11): 245-249.

Krumpál, M. & D. Cyprich. On the flea (Siphonaptera) fauna in birds' (Aves) nests in the protected landscape area of Ponitrie. Rosalia (Nitra) 7: 213-222.

(1991)

Lindsay, L. R., I. K. Barker, G. A. Surgeoner, S. A. McEwen, L. A. Elliot & J. Kolar. Apparent incompetence of *Dermacentor variabilis* (Acari: Ixodidae) and fleas (Insecta: Siphonaptera) as vectors of *Borrelia burgdorferi* in an *Ixodes dammini* endemic area of Ontario, Canada. J. med. Ent. 28(5): 750-753.

Medvedev, S. G., M. K. Stanjukovich, M. P. Tiunov & G. V. Farafonova. Ectoparasites of bats from the Far-East of the USSR. Parazitologiya 25(1): 27-38.

Méndez, E. Fleas (Siphonaptera) of the Sian Ka'an Biosphere Reserve, Quintana Roo, México. Chapter 20. In: Diversidad Biológica en la Reserva de la Biosfera de Sian Ka'an, Quintana Roo, México. pp. 281-306.

Moller, A. P. Ectoparasite loads affect optimal clutch size in swallows. Functional Ecol. 5(3): 351-359.

Moser, B. A., P. G. Koehler & R. S. Patterson. Separation of cat flea (Siphonaptera: Pulicidae) instars by individual rearing and head width measurements. J. econ. Ent. 84(3): 922-926.

Moser, B. A., P. G. Koehler & R. S. Patterson. Effect of larval diet on cat flea (Siphonaptera: Pulicidae) development times and adult emergence. J. econ. Ent. 84(4): 1257-1261.

Navarro-Sigüenza, A. G. & J. E. Llorente-Bousquets. Museos, colecciones biológicas y la conservación de la biodiversidad: una perspectiva para México. *Memorias del Seminario sobre Conservación de la Diversidad Biologica de México.* No. 3(1991): 1-31.

Patton, J. Flea control from a veterinarian's point of view. Pest Management 10(6): 25-28.

Peace, T. Minimizing flea problems in your clinic. Vet. Med. 86(8): 780.

Pilgrim, R. L. C. Presidential address - Fleas. New Zealand Ent. 14: 1-9.

(1991)

Pilgrim, R. L. C. External morphology of flea larvae (Siphonaptera) and its significance in taxonomy. Fla Ent. 74(3): 386-395.

Plant, J. D. Recognizing the manifestations of flea allergy in cats. Vet. Med. 86(5): 482-487.

Poulin, R. Group-living and infestation by ectoparasites in passerines. Condor 93(2): 418-423.

Rothschild, M. Arrangement of sperm within the spermatheca of fleas, with remarks on sperm displacement. Biol. J. Linn. Soc. 43: 313-323, figs 1-12.

Rybin, S. N. A new flea species from south Kirghizia, *Nycteridopsylla singula* sp. n. (Siphonaptera: Ischnopsyllidae). Parazitologiya 25(2) 172.

Samuel, W. M., D. A. Welch & B. L. Smith. Ectoparasites from elk (*Cervus elaphus nelsoni*) from Wyoming. J. Wildl. Dis. 27(3): 446-451.

Scharf, W. C. Geographic distribution of Siphonaptera collected from small mammals on Lake Michigan islands. Gt Lakes Ent. 24(1): 39-44.

Vedmed, A. I., S. V. Krylova & I. V. Issi. The microsporidium *Pulicispora xenopsyllae* gen. n., sp. n. from fleas of the genus *Xenopsylla*. Parazitologiya 25(1): 13-19.

Williams, B. Indicators of sexual maturation and regression in the female cat flea, *Ctenocephalides felis*. Med. vet. Ent. 5(3): 369-376, figs. 1-5.

Wisselink, M. A., J. P. Koeman, T. S. G. A. M. Vandeningh & A. Willemse. Investigations on the role of flea antigen in the pathogenesis of German Shepherd dog Pyoderma (GSO). Tijdschr. voor Diergeneeskunde 116(10): 513-520.

Xie Bao-qi, Fang Zhu & Hu Gui. A study on the classification of *Neopsylla biseta* in China (Siphonaptera: Hystrichopsyllidae). Acta zootax. sin. 16(2): 232-239, figs. 1-16. [*Neopsylla biseta eleusina* (Li, 1980) comb. n. &

(1991)

Pilgrim, R. L. C. External morphology of flea larvae (Siphonaptera) and its significance in taxonomy. Fla Ent. 74(3): 386-395.

Plant, J. D. Recognizing the manifestations of flea allergy in cats. Vet. Med. 86(5): 482-487.

Poulin, R. Group-living and infestation by ectoparasites in passerines. Condor 93(2): 418-423.

Rothschild, M. Arrangement of sperm within the spermatheca of fleas, with remarks on sperm displacement. Biol. J. Linn. Soc. 43: 313-323, figs 1-12.

Rybin, S. N. A new flea species from south Kirghizia, *Nycteridopsylla singula* sp. n. (Siphonaptera: Ischnopsyllidae). Parazitologiya 25(2) 172.

Samuel, W. M., D. A. Welch & B. L. Smith. Ectoparasites from elk (*Cervus elaphus nelsoni*) from Wyoming. J. Wldlf. Dis. 27(3): 446-451.

Scharf, W. C. Geographic distribution of Siphonaptera collected from small mammals on Lake Michigan islands. Gt Lakes Ent. 24(1): 39-44.

Vedmed, A. I., S. V. Krylova & I. V. Issi. The microsporidium *Pulicispora xenopsyllae* gen. n., sp. n. from fleas of the genus *Xenopsylla*. Parazitologiya 25(1): 13-19.

Williams, B. Indicators of sexual maturation and regression in the female cat flea, *Ctenocephalides felis*. Med. vet. Ent. 5(3): 369-376, figs. 1-5.

Wisselink, M. A., J. P. Koeman, T. S. G. A. M. Vandeningh & A. Willemse. Investigations on the role of flea antigen in the pathogenesis of German Shepherd dog Pyoderma (GSO). Tijdschr. voor Diergeneeskunde 116(10): 513-520.

Xie Bao-qi, Fang Zhu & Hu Gui. A study on the classification of *Neopsylla biseta* in China (Siphonaptera: Hystrichopsyllidae). Acta zootax. sin. 16(2): 232-239, figs. 1-16. [*Neopsylla biseta eleusina* (Li, 1980) comb. n. &

(1991)

Neopsylla biseta bijiangensis Xie, Fang & Hu, 1991]

Xie Bao-qi, Yang Kuang-rong, Tian Jie & Xie Shu. Descriptions of new species and subspecies of *Doratopsylla* from Yunnan Province (Siphonaptera: Hystrichopsyllidae). Acta zootax. sin. 16(2): 240-247, figs. 1-20. [*Doratopsylla ji* Xie & Tian, 1991, *Doratopsylla liui* Xie & Xie, 1991 & *Doratopsylla coreana jianchuanensis* Xie & Yang, 1991]

It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

"Notwithstanding the inconveniences attending this little insect, there is something pleasing in the appearance of the flea. Its motions are elegant, and all its postures indicate agility. The shell with which it is enveloped is in a state of perpetual cleanliness, while the muscular power which it is capable of exerting is so extraordinary, as to excite our wonder at so much strength confined and concentrated within so small a space; this species being able to spring, on the most moderate computation, to the distance of at least two hundred times its own length, and drag a weight eight times heavier than itself. It has sometimes become a favorite with ladies, who have pleased themselves with keeping, taming, and feeding it. A golden chain has been made for it with a lock and key; and being kept in a box with wool, in a warm place, and fed daily, it has been known to live for six years.

"The *Pulices* of birds and of mammalia ought to be most carefully examined. There are a vast number of species which have been confounded with *P. irritans*."
(Unattributed quote from : Samouelle, G. 1819. *The Entomologist's Useful Compendium; or an introduction to the knowledge of British insects...* Thomas Boys, London. p. 234.)

▲★▲★▲★▲★▲

**Best wishes from Robert E. & Joanne H. Lewis for a joyous
holiday season and the happiest of New Years**



Flea News

Department of Entomology
Iowa State University
Ames, IA 50011-3222 USA

Cpt. Michael W. Hastriter
603 Beretta Way
Bel Air, MD 21014

FIRST CLASS

Department of Entomology - Iowa State University - Ames, IA 50011-3222
Tel. (515) 294 1815 Fax. (515) 294 8027 BitNet A1.REL@ISUMVS